

WILLOW RUN, MI, USA

WMO: 725376

Lat: 42.233N

Lon: 83.533W

Elev: 777

StdP: 14.29

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14853

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.6	5.8	-8.0	3.7	1.5	-2.2	5.0	7.6	27.9	23.2	25.5	28.1	8.2	220	0.556

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.3	92.0	73.9	89.2	72.6	86.3	71.0	77.1	87.6	75.2	84.7	73.5	82.3	11.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.6	128.9	82.9	72.3	122.9	81.0	70.5	115.5	79.0	40.9	87.1	39.1	85.0	37.6	82.7	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.7	21.3	19.0	DB	-6.1	95.8	7.5	3.6	-11.5	98.3	-15.9	100.4	-20.1	102.4	-25.6	105.0	(4)
			WB	-6.8	80.2	7.3	1.9	-12.0	81.6	-16.3	82.7	-20.4	83.8	-25.7	85.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.3	25.0	27.1	36.6	48.6	60.1	69.8	73.6	71.7	64.7	52.8	40.9	30.9
	DBStd	19.23	11.42	10.63	10.62	8.80	8.67	6.86	5.94	5.51	7.49	8.41	8.93	9.82
	HDD50	2955	775	642	438	127	13	0	0	0	2	65	296	597
	HDD65	6194	1239	1060	882	495	205	32	5	9	96	390	723	1058
	CDD50	3055	1	2	22	86	324	593	731	674	443	151	24	4
	CDD65	819	0	0	1	4	52	175	271	218	87	11	0	0
	CDH74	7866	0	0	14	69	604	1688	2699	1893	795	102	2	0
	CDH80	2757	0	0	3	10	194	623	1046	612	251	18	0	0
Wind	WSAvg	9.0	10.6	10.4	10.2	10.4	8.6	7.7	7.2	6.9	7.2	8.8	9.8	10.0
Precipitation	PrecAvg	32.2	2.1	1.8	2.1	2.8	3.7	3.4	3.3	3.3	2.9	2.5	2.3	2.1
	PrecMax	41.5	4.4	4.1	4.2	5.6	7.3	6.4	6.6	6.4	5.8	6.8	5.5	4.4
	PrecMin	26.1	0.6	0.6	0.4	0.4	0.9	1.2	1.5	0.4	0.9	0.0	0.7	0.6
	PrecStd	4.1	1.0	0.8	1.1	1.2	1.8	1.3	1.3	1.5	1.4	1.3	1.1	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.4	61.0	74.7	81.3	90.2	94.7	96.7	93.3	92.2	82.3	69.3	61.3
		MCWB	54.3	53.5	62.0	64.0	70.2	74.5	76.7	74.8	73.8	68.5	57.0	56.9
	2%	DB	51.5	52.3	65.8	74.9	85.4	90.4	92.4	89.2	86.9	76.5	63.3	54.4
		MCWB	48.1	46.9	54.9	59.6	68.2	72.4	74.5	73.7	70.6	64.4	55.3	50.3
	5%	DB	45.3	45.9	59.3	70.0	81.1	87.0	89.3	86.1	82.0	72.0	59.7	48.8
		MCWB	41.6	40.7	51.1	56.8	66.4	70.7	73.3	72.0	68.7	61.6	53.7	44.8
	10%	DB	39.6	41.0	53.6	64.6	76.2	83.2	85.9	83.3	78.6	67.2	54.9	43.8
		MCWB	36.7	37.0	46.8	53.6	63.9	69.0	71.2	70.6	66.6	59.0	49.3	39.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.1	54.9	62.8	66.7	73.2	77.3	79.8	78.7	76.8	71.1	61.0	57.0
		MCDB	56.6	58.5	74.1	78.4	84.9	90.3	91.2	89.3	88.0	80.0	65.0	61.3
	2%	WB	47.1	47.2	57.0	62.2	70.8	75.0	77.4	76.5	73.4	66.6	57.4	50.8
		MCDB	49.1	52.2	63.1	70.5	81.9	86.5	88.4	85.1	81.7	73.0	61.7	54.0
	5%	WB	40.8	40.9	52.6	58.7	68.4	73.1	75.4	74.5	71.1	63.5	53.8	45.2
		MCDB	43.8	45.5	58.7	67.9	78.1	83.5	84.9	82.3	78.6	69.7	58.7	48.8
	10%	WB	36.5	37.1	46.3	54.9	65.7	71.2	73.5	72.7	68.7	59.9	49.9	40.0
		MCDB	39.1	40.6	52.6	63.8	74.5	80.2	82.2	80.3	75.7	66.1	54.9	43.6
Mean Daily Temperature Range	5% DB	MDBR	13.8	15.4	18.1	21.3	21.7	21.2	21.3	20.3	21.8	19.3	15.8	12.5
		MCDBR	17.5	20.8	27.2	29.4	26.7	25.3	23.8	22.7	26.1	25.2	22.0	17.5
		MCWBR	15.3	16.8	18.0	17.5	13.8	11.6	10.3	10.4	12.8	14.6	16.4	14.4
	5% WB	MCDBR	16.8	20.2	25.1	25.7	23.6	21.6	20.3	19.1	21.5	21.8	19.7	16.4
		MCWBR	15.2	16.8	18.2	17.4	13.9	11.6	10.4	10.2	11.8	14.0	16.2	14.7
Clear-Sky Solar Irradiance	taub	0.299	0.337	0.354	0.401	0.431	0.450	0.441	0.433	0.388	0.356	0.317	0.311	
	taud	2.393	2.264	2.337	2.227	2.194	2.179	2.227	2.239	2.344	2.417	2.489	2.392	
	Ebn at Noon	268	272	282	275	269	263	264	262	267	263	260	251	
	Edh at Noon	25	34	36	43	45	46	43	42	35	29	23	23	
All-Sky Solar Radiation	RadAvg	490	770	1118	1462	1704	1916	1929	1671	1354	883	554	388	
	RadStd	44	97	112	128	149	123	97	64	119	76	73	33	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+1.58	N/A	N/A	+1.37	N/A	N/A	N/A	-410	+380	+229
Regional (63 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+149	N/A

Nomenclature: See separate page